



Nonlinear analysis of damped semi-rigid frames considering moment–shear interaction of connections

Reza Attarnejad ^{a,b}, Akbar Pirmoz ^{a,*}

^a School of Civil Engineering, University of Tehran, Iran

^b The International Center for Numerical Methods in Engineering, Spain

ARTICLE INFO

Article history:

Received 11 July 2013

Received in revised form

15 January 2014

Accepted 13 February 2014

Available online 25 February 2014

Keywords:

Semi-rigid frames

Nonlinear vibration analysis

Flexible connections

Moment–shear interaction

Finite element method

ABSTRACT

Nonlinear vibration analysis of semi-rigid frames, considering moment and shear interaction (MVI) for the joint, is performed in this article. The numerical model of the semi-rigid frames is created using the nonlinear finite element (FE) formulation. In the numerical model, the beams and the columns are formulated based on the Euler–Bernoulli beam theory and the Partially Restrained (PR) connections are simulated as a discrete rotational springs and a parallel rotational damper. The previously formulated three-parameter model is modified and the nonlinear moment–rotation relation of the connection is tied to its shear force to assess the interaction effects of the moment and shear of the connections. An analysis algorithm is proposed to solve the eigenvalue problem of the semi-rigid frames subjected to pre-loading conditions, taking into account the MVI effects. The method is verified through comparison with analytical examples.

© 2014 Elsevier Ltd. All rights reserved.

1. Introduction

Although there is a considerable research on the mechanical/structural characteristics of the semi-rigid frames, a less contribution deals with dynamic response analysis of the pre-loaded nonlinear semi-rigid frames. Linear dynamic response analysis of damped semi-rigid frames is performed by Kawashima and Fujimoto [1]. The results of their study showed the impact of the dynamic characteristics of the flexible connections on the vibration response of semi-rigid frames. Chan [2] developed a numerical model for vibration analysis of the semi-rigid frames with elastic connections. The method later was extended and used in the seismic analysis of semi-rigid frame with nonlinear connections [3,4]. Lui and Lopes [5] studied the vibration response of semi-rigid frames, concluding the sensitivity of their response to connection behavior and P–Δ effects. Xu and Zhang [6] studied the dynamic response of semi-rigid frames with rotational dampers at the connections. Their study showed that there is an optimal damping for a given system, at which the seismic response of the frame is minimum. Sekulovic et al. [7] analyzed the dynamic response of the semi-rigid frames considering the connection nonlinearity and damping. Dynamic analysis of two frames with different connection types showed the crucial rule of the semi-rigid beam column connections on the stability and seismic response of these frames. Sophianopoulos [8] studied the joint

flexibility effect on the free vibration of elastic steel frames. The results of this work showed the considerable impact of connection response on the dynamic characteristics of semi-rigid frames. Seismic performance assessment of semi-rigid frames by using nonlinear dynamic analysis of their FE models showed the superiority of these frames to the corresponding rigid moment frames [9,10]. Da Silva et al. [11] studied the resonance response of a semi-rigid frame considering the nonlinear connection behavior. The analysis results showed that resonance does not occur in nonlinear models due to energy dissipation nature of the nonlinear connections. In reference [12] vibration and instability of semi-rigid frames with elastic joint stiffness is studied subjected to pre-loading conditions. The results of this study show the impact of connection flexibility and pre-loading conditions on the dynamic response of the slender flexible frames.

In all these studies, the rotational response of the connections is considered ignoring the interaction of the joint shear on the rotational characteristics of the semi-rigid connections and dynamic response analysis of these systems considering MVI is an unresolved issue. Herein, the eigenvalue analysis of the pre-loaded semi-rigid frames is performed considering connection nonlinearity, MVI and damping.

2. Basic assumptions

The finite element (FE) modeling of the semi-rigid frames considering MVI and establishing the new algorithm for vibration

* Corresponding author. Tel.: +98 2144854516; fax: +98 66403808.

E-mail address: a.pirmoz@ut.ac.ir (A. Pirmoz).

response analysis of these frames are based on the following assumptions:

- The prismatic beams and columns are modeled elastic according to the Euler beam theory.
- Beam-to-column PR connections are modeled as discrete zero-length joints and the connection eccentricity is ignored.
- Hysteretic damping of the PR connections is captured through a nonlinear moment–rotation relation for the joint (nonlinear rotational spring) and a linear zero-length rotational damper is defined to add additional viscous damping to the system. Hence, the nonlinear rotational spring and rotational damper act in parallel.
- The first nonlinearity source of the model is the constitutive law defined for the nonlinear rotational response of the joints (PR connections).
- The second source of nonlinearity is the relationship between the rotational stiffness of the connection and its shear.
- The implemented equations to relate the rotational response of the connection to its shear are applicable for bolted top-seat and web angle(s) connection, TSWA.
- In the free vibration response analysis, the nodal displacements due to initial pre-loading are small and the geometric nonlinearity is ignored.
- The mass source includes distributed element mass and lumped mass on the structural nodes.
- The characteristics of the viscous rotational dampers do not change within analysis.

3. Formulation of the beams and columns

Beam/column elements are formulated according to the Euler beam theory ignoring the shear deformations. The corresponding stiffness matrix, K_b , is in the form of Eq. (1).

$$K_b = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} \\ & & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & -\frac{2EI}{L} \\ & & & \frac{EA}{L} & 0 & 0 \\ sym. & & & & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ & & & & & \frac{4EI}{L} \end{bmatrix} \quad (1)$$

in which E , I , A and L are the modulus of elasticity, moment of inertia, sectional area and the length of the member, respectively.

4. Modeling of the flexible connections

Taking into account the rotational flexibility of the PR connections can be achieved either modifying the flexural stiffness of the beam or modeling the connection as a discrete joint element. A two-node zero-length element with three degrees of freedom (DOF) for each node is implemented to model the flexible connections in this study. This element is placed between the coincident end nodes of beam and column members/components (nodes i and j shown in Fig. 1-a). A rotational damper is also defined between these nodes as shown in Fig. 1-a. The stiffness matrix of the joint, K_{joint} , is in the form of Eq. (2).

$$K_{joint} = \begin{bmatrix} k_a & 0 & 0 & -k_a & 0 & 0 \\ & k_s & 0 & 0 & -k_s & 0 \\ & & k_T & 0 & 0 & -k_T \\ & & & k_a & 0 & 0 \\ sym. & & & & k_s & 0 \\ & & & & & k_T \end{bmatrix} \quad (2)$$

Where, k_a and k_s denote the elastic axial and shear stiffness of the connection. The nonlinear moment–rotation (M – θ) response of the joint is defined by Eq. (3), according to the previously defined three-parameter power model [13]. The tangent rotational stiffness of the connection, k_T , is differential of the connection moment (M) with respect to its rotation (θ) and calculated by using Eq. (4). Fig. 1-b) schematically shows the connection response and its mechanical characteristics. Herein, the original three-parameter equation is modified to consider the effects of the joint shear force by using R_{stiff} , the reduction factor for the rotational stiffness of the connection.

$$M = \frac{k_0^* \theta}{[1 + (\theta/\theta_0)^n]^{(1/n)}} \quad (3)$$

$$k_T = \frac{dM}{d\theta} = \frac{k_0^*}{[1 + (\theta/\theta_0)^n]^{(1/n)}} \left(1 - \frac{(\theta/\theta_0)^n}{1 + (\theta/\theta_0)^n} \right) \quad (4)$$

$$\theta_0 = \frac{M_u}{k_0} \quad (5)$$

$$k_0^* = R_{stiff} \times k_0 \quad (6)$$

as stated, M and θ in the above equations are the applied moment and the corresponding rotation and θ_0 is a reference rotation defined by Eq. (5). Also, M_u , n and k_0 are the ultimate moment capacity of the connection, a shape parameter and the initial rotational stiffness of the connection, respectively. k_0^* is the reduced initial rotational stiffness of the connection due to existence of shear force in the connection. Details on calculating the reduction factor for the initial rotational stiffness, R_{stiff} , are discussed in the following section.

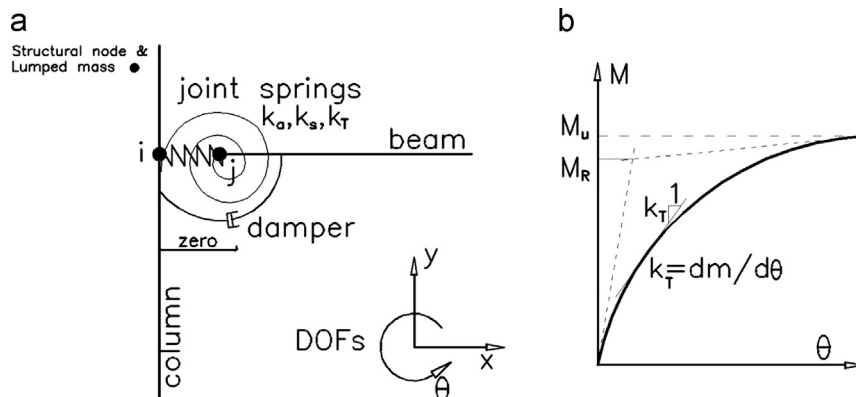


Fig. 1. Modeling of the semi-rigid connections (a) and nonlinear moment-rotation response of the PR connections (b).

5. Moment–shear interaction (MVI)

Almost all types of the semi-rigid connections are primarily designed to carry the shear forces of the attached beam. Even in the cases of their intentional design as a rotational element, the connections may sustain some shear and axial forces [14]. The previous numerical studies on the interaction effects of these internal forces on the mechanical response of bolted angle connections have shown that this interaction can have a considerable effect on the $M-\theta$ response of these PR connections [15–19], and accordingly on the frame response.

In spite of this, there is a little work on the MVI of the semi-rigid frames and the effects of shear forces on the rotational response of the connections are often ignored in the frame analyses.

By using nonlinear FE method, Pirmoz et al. [16] investigated the effects of the MVI on the rotational response of the bolted angle connections. They proposed a formulation (Eq. (7)) to modify the initial rotational stiffness of this type of bolted connection subjected to a constant initial shear force:

$$R_{stiff} = 1 - \left(\frac{S_R}{24W_y} \right) S_n^2 \quad (7)$$

$$S_n = \frac{S_C}{S_R} \quad (8)$$

where, R_{stiff} denotes the reduction factor for the initial rotational stiffness, S_R is the corresponding shear value at M_R , S_C is the applied shear force on the connection and W_y is the ultimate shear strength of the web angle(s) increased by 1.67. M_R is the reference moment defined in Fig. 1-b.

6. Dynamic formulation and vibration analysis

Free vibration response of a structural system can be examined through the eigenvalues of its general dynamic equation (Eq. (9)).

$$M\ddot{U} + C\dot{U} + KU = 0 \quad (9)$$

where, K is (tangential) stiffness matrix of the frame and U is the N -dimensional vector of the nodal displacements and its first and second derivatives, $\dot{U}$ and $\ddot{U}$, are coupled with the mass matrix (M) and the damping matrix (C) respectively. In order to formulate the dynamic equation of a semi-rigid frame, connections are assumed as mass-less elements while the beams and the columns define the mass matrix of the frame.

Because of the inherent viscous damping of the semi-rigid connections, considered in the frame analysis, the system is non-classically damped and the orthogonality condition of natural modes does not apply. Accordingly, the well known QR method is implemented to solve the eigenvalue problem of Eq. (9). In this method, the general dynamic equation (Eq. (9)) is replaced with:

$$A\dot{Y} + BY = 0 \quad (10)$$

where, A and B are $2N$ -dimensional matrices:

$$A = \begin{bmatrix} 0 & M \\ M & C \end{bmatrix}, B = \begin{bmatrix} -M & 0 \\ 0 & K \end{bmatrix} \text{ and } Y = \begin{Bmatrix} \dot{U} \\ U \end{Bmatrix} \quad (11)$$

hence, the corresponding eigenvalue problem becomes:

$$(sA + B)\phi = 0 \quad (12)$$

For an underdamped system, the eigenvalues (s_j) and eigenvectors (ϕ_j) are in complex conjugate pairs—signed with superscript $*$ in the following equation—with zero or negative real parts. In this case,

the eigenvalues of the j th mode will be:

$$s_j = s_{j+N}^* = -\omega_j \xi_j + i\omega_{dj}, \quad (j = 1, 2, \dots, N), \quad (13)$$

$$\omega_j = |s_j|, \quad \xi_j = -R_e(s_j)/|s_j| \text{ and } \omega_{dj} = \omega_j \sqrt{1 - \xi_j^2} \quad (14)$$

where, ω_j , ω_{dj} , ξ_j and j are the modal frequency, the damped modal frequency, the modal damping ratio and node number, respectively [20].

7. Loading pattern and the solution algorithm

In order to assess the MVI impact on the vibration response of the damped semi-rigid frames, the connections should be loaded before the vibration analysis. Because of this, the eigenproblem solution involves two analysis phases, the nonlinear static analysis and the frequency analysis. The first analysis needs a nonlinear solution for the system under a static loading:

$$F = UK \quad (15)$$

where, F is the nodal force vector and K and U are previously defined (tangent) stiffness and the nodal displacement vector of the system. At the beginning of the analysis, the stiffness matrix is calculated and the nodal displacement vector is calculated solving Eq. (15):

$$U^1 = (K^1)^{-1}F \quad (16)$$

The nodal displacement vector, U , is updated iteratively (Eqs. (17) and (18)) solving against the residual force vector, ψ .

$$U^{r+1} = U^r + \Delta U^{r+1} \quad (17)$$

$$\Delta U^{r+1} = (K^r)^{-1}\psi^r \quad (18)$$

where, r is the iteration number, U^{r+1} is the nodal displacement vector for the $(r+1)$ th iteration and ΔU^{r+1} is the incremental nodal displacement vector for the $(r+1)$ th iteration. After calculation of ΔU , the stiffness matrix is updated according to Eq. (4) and the residual force vector, ψ , is calculated (Eq. (19)) and examined for convergence conditions (Eq. (20)):

$$\psi^r = K^r \Delta U^r + F \quad (19)$$

$$\frac{\sqrt{\sum_{i=1}^N (\psi_i^r)^2}}{\sqrt{\sum_{i=1}^N (f_i^r)^2}} \times 100 \leq \epsilon \quad (20)$$

where, N is number of the structural nodes, i denotes the node number and ϵ is a very small number as the convergence tolerance. Also, ψ_i^r and f_i^r are unbalanced internal and external force of the i th node at the r th iteration, respectively.

Solution of Eq. (15) should satisfy two different types of convergence criteria. The first criterion is the traditional criterion (Eq. (18)) that controls the unbalanced (residual) forces, ψ , due to the nonlinear $M-\theta$ response of the connections (Fig. 2). The term “exact solution” shown in Fig. 2 is defined by Eq. (3) for the nonlinear joint elements.

The second convergence criterion, however, assesses R_{stiff} . R_{stiff} is a function of the joint shear force obtained from the previous converged iteration. At the beginning of the nonlinear static analysis, Eq. (15) is solved assigning a unit value for R_{stiff} . After achieving the convergence based on the residual force criterion through satisfying Eq. (20), the nodal forces are calculated and R_{stiff} is updated according to Eq. (7). Then Eq. (15) is reanalyzed. This cycle is iterated until the difference between two successive R_{stiff} diminishes. Fig. 3 illustrates this algorithm, programmed in FORTRAN, for the vibration analysis of the semi-rigid frames.

8. Verification of the method

Two simple problems are solved analytically and the obtained results are compared with the results of the program, to assess the validity of the developed program.

8.1. Free vibration of a dashpot

Vibration frequency of a simple dashpot, consisting of a lumped mass supported by two springs and a damper (in parallel), is examined. This system is solved as a Single-Degree-of-Freedom (SDOF) oscillator, analytically and by using the program. Table 1

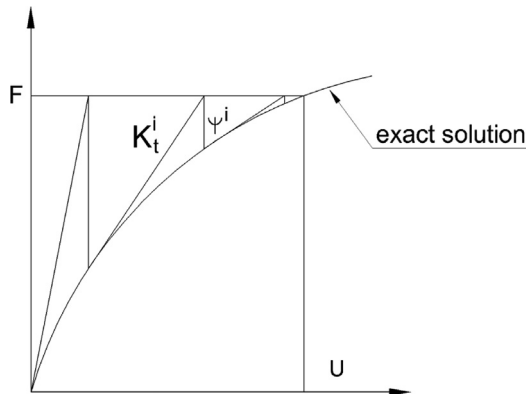


Fig. 2. Tangential stiffness solution algorithm and residual force vector of the i th iteration, ψ^i .

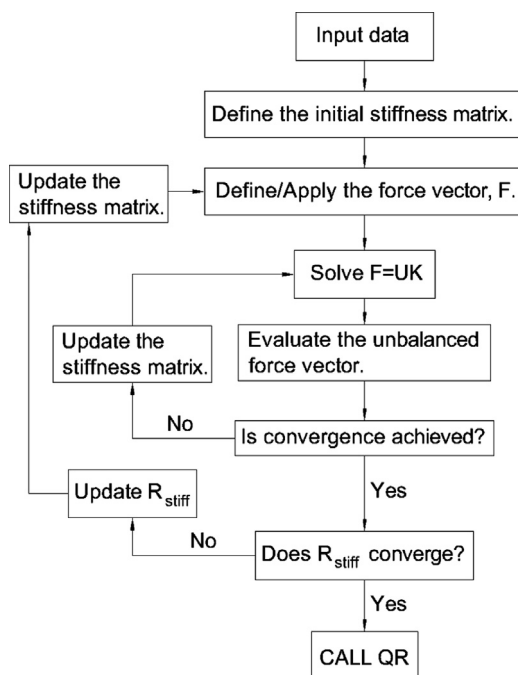


Fig. 3. The programmed algorithm for the eigenproblem solution.

compares the results obtained through both methods. As seen in this table, the results obtained by the program are agreed well with the exact solution.

In this table, m is the lumped mass of the oscillator, $k_{1,2}$ are the springs' stiffness, c is the damper damping ratio, ξ denotes the damping ratio and ω_n and ω_D are the natural and damped circular frequency, respectively.

8.2. Simply supported beam with a flexible connection

The main object of this research is to evaluate the impact of the moment–shear interaction of the semi-rigid joints on the vibration response of the frame. Since there is not any experimental test or numerical study in this regard, an analytical example is solved to verify the method.

A simple beam with a semi-rigid connection (Fig. 4) is iteratively solved. This simple system is statically indeterminate and the asymmetric support conditions tie the support reactions to the stiffness of the rotational spring (degree of fixity of the flexible end). On the other hand, the rotational stiffness of the spring is a function of its shear (internal) force, according to Eqs. (4–7). By using traditional structural analysis methods, the moment, M_s , and shear, V_s , at the semi-rigid connection of the beam are obtained according to Eqs., (21) and (22) when the beam is loaded at the mid-span.

$$M_s = \frac{PL^2}{(16EI/K_s) + (16L/3)} \quad (21)$$

$$V_s = \frac{M_s}{L} + 0.5P \quad (22)$$

in the above equations, P is the applied concentric load, L is the beam length, E and L are the modulus of elasticity and the length of the beam, respectively, and K_s is the rotational stiffness of the attached spring.

Table 2 presents the results of the analytical computation for the five iterations performed to achieve convergence. From this table, the FE program can accurately capture the exact solution for the static and vibration problems considering MVI effects.

9. Analyses of the models

The vibration characteristics of a single-storey and a three-storey frame are examined considering MVI and damping of the connections. Fig. 5 shows the layout and loading of these frames.

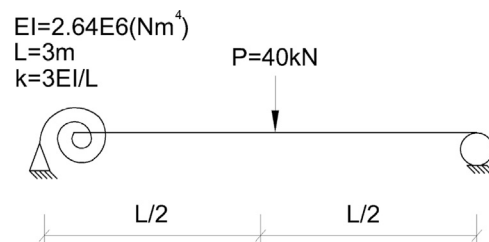


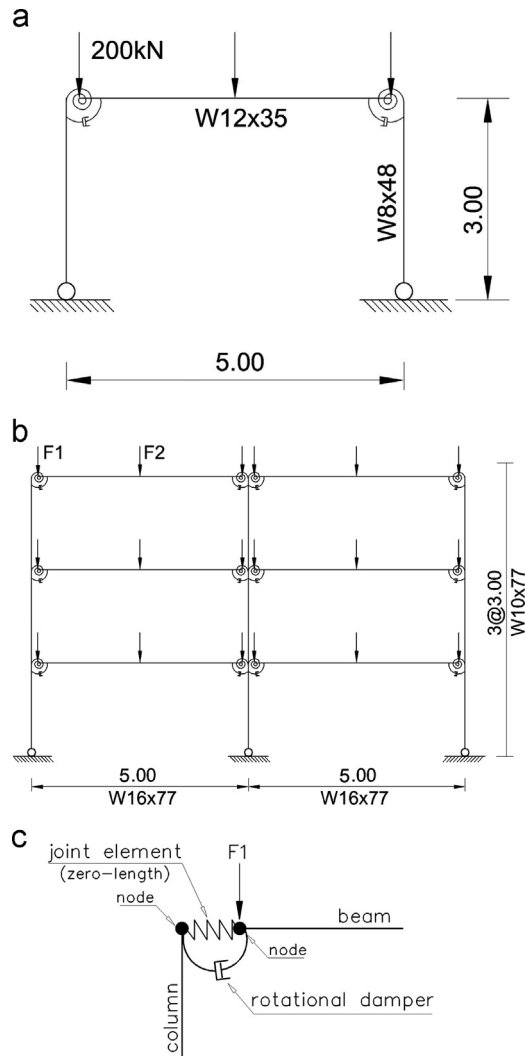
Fig. 4. The simple beam with a parallel rotational spring and damper.

Table 1
accuracy examination results for the SDOF oscillator.

$m = 1.5e5(\text{kg})$	$\omega_n = \sqrt{\frac{k_1 + k_2}{m}}$	$\xi = \frac{c}{2\sqrt{mk}}$	$\omega_D = \omega_n \sqrt{1 - \xi^2}$	Program		
$k_1 = 100(\text{kN/m})$				ω_n	ξ	ω_D
$k_2 = 10(\text{kN/m})$						
$c = 100(\text{kN s/m})$	0.8563488	0.3892494	0.7888107	0.8563489	0.3892494	0.7888107
$c = 200(\text{kN s/m})$	0.8563488	0.7784988	0.5374840	0.8563488	0.7784989	0.5374838

Table 2Analytical computations for a beam with an elastic semi-rigid connection, $W_y = 41.75$ kN, $S_R = 3.8$ kN ∞ .

Iter.	R_{stiff} (Eq. (7))	R_{stiff} (FE program)	$K_s = R_{stiff} \times k$	M_s (Nm)	V_s (N)	S_n	$\theta = M_s/K_s$ (mrad)	θ FE program (mrad)
1	1.00	1.00	2640000	11250.00	23750.00	6.250	4.2614	4.2574
2	0.85186	0.85190	2248907.2	10350.05	23450.02	6.171	4.6023	4.5979
3	0.85558	0.85562	2258724.5	10374.39	23458.13	6.173	4.5930	4.5887
4	0.85548	0.85552	2258460.6	10373.74	23457.91	6.173	4.5933	4.5889
5	0.85548	0.85552	2258467.6	10373.76	23457.92	6.173	4.5933	4.5889

**Fig. 5.** a: single-storey frame, b: three-storey frame (beams are $W10 \times 77$ and column are $W16 \times 77$), c: beam-to-column connection assembly, Fig. 5 The analyzed frames geometry and loading.

Beam-to-column connections of these frames are bolted TSWA connections (Fig. 6 and Table 3). In designing of the connections, it is assumed that the web angles sustain the half of the applied shear force and the remaining shear force is carried by the top and seat angles, especially due to the axial stiffness of the outstanding vertical leg of the top angle connected to the column [17].

The three parameters of the implemented power-model (M_i , K_i and n) for the M – θ relation are estimated using the proposed equations by Kishi and Chen [13]. The modification factors of Eqs. (7) and (8) are defined based on the analysis of the semi-rigid connection with a cantilevered beam. However, the span length is considered in the case of a full frame. Fig. 6 presents the geometric parameters of the connections for these frames. For

the single-storey frame, the reference shear for the connections, S_R , is 30 kN, the assumed rotational damper damping coefficient is 1000 kN m/s/rad. and a lumped mass of 12×10^3 kg is considered at all structural nodes.

For the three-storey frame, S_R is 55 kN, W_y is 1372 kN and a lumped mass of 45×10^3 (kg) is considered on all structural nodes of the frame.

10. Structural models

Each beam is modeled using two elastic Euler beam elements and the columns are modeled as a single beam element. To achieve this, three nodes are defined for each beam (one in both ends and one in the middle) and two nodes are defined for each column. The end nodes of the beams and columns are coincident and joined together using the zero-length connection element shown in Fig. 1-a), to simulate the PR connections. In this way, the single-storey frame (Fig. 5-a) includes 7 nodes, 4 elastic Euler elements (beams/columns), two joint elements and two rotational dampers. A similar modeling pattern is followed for the three storey frame. As a result, this frame has 30 nodes, 21 elastic elements (beams/columns), 12 joint elements and 12 rotational dampers (Fig. 5-b).

Three types of connection models are considered in analyzing of the frames:

- 1- Elastic connection response (Type I) in which the rotational response of the connection is kept unchanged during the loading and analysis.
- 2- Full nonlinear connection response (Type II) in which the moment-rotation relation of the connection is nonlinear (Eq. 3) and the tangent stiffness is used for the Eigensolution.
- 3- Full nonlinear connection response with MVI effects (Type II) in which the nonlinear moment-rotation relation of the connections is modified considering the connection shear as discussed previously.

The gravity loading is applied as concentrated loads on the beam nodes (Fig. 5-c).

11. Single-storey frame response

Vibration analysis of this frame with different joint types showed that its dynamic response can be considerably affected by the joint response. Although assuming an elastic response simplifies the analysis, it results some error as a penalty for this simplification. A similar applies to ignoring the effects of the joint shear. Table 4 lists the obtained frequencies for the first mode of vibration for three connection types.

From this table, when the joint response is elastic, the frequency of vibration is 1.267 times the vibration frequency for the nonlinear response and 1.743 times the vibration response for the nonlinear joint with MVI effects. This table also shows that

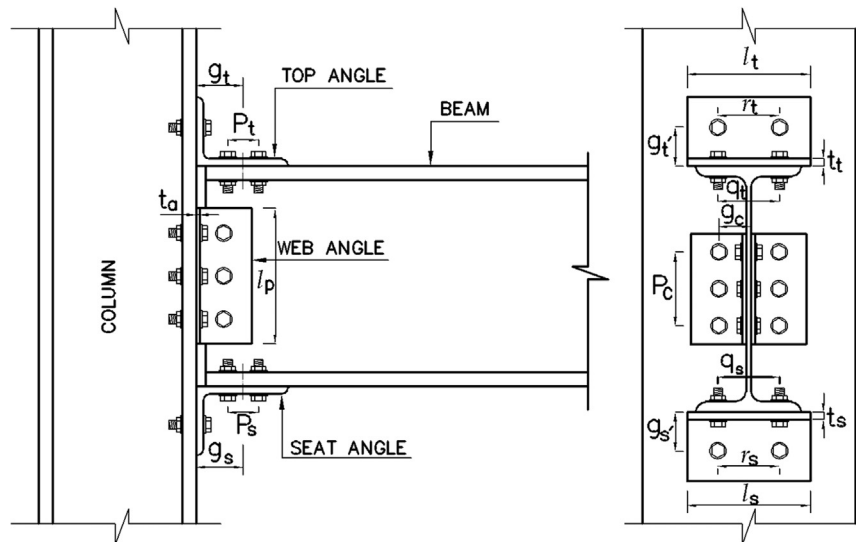


Fig. 6. Detail parameters for TSWA connections of the frames.

Table 3
Geometric characteristic of the connection (mm).

	Top-seat angles	Web angles	l_{bs}	$g_{t's'}$	g_{ts}	p_{bs}	q_{ts}	r_{ts}	p_{bc}	g_c	l_p	M_u (kNm)	K_t (kNm)	n
single-storey	$L6 \times 4 \times 7/16$	$L4 \times 4 \times 5/16$	170	55	55	60	90	90	50	50	120	160	49400	0.8
three-storey	$L6 \times 4 \times 7/16$	$L4 \times 4 \times 5/16$	200	55	55	60	90	90	50	50	150	240	72000	1.0

Table 4
Vibration frequency of the first mode of vibration for the single-storey frame, $W_y = 225$ (kN), $S_R = 36$ (kN).

Connection type	$C = 1000$ (kN m s/rad)				$C = 0.0$ (kN m s/rad)			
	ω_n	ξ	ω_D	Period (s)	ω_n	ξ	ω_D	Period (s)
I	5.314	0.009	5.314	1.182	5.309	0.0	5.309	1.182
II	4.216	0.101	4.194	1.490	4.123	0.0	4.123	1.490
III	3.224	0.325	3.049	1.949	2.962	0.0	2.962	2.121

the connection response has a considerable effect on the modal damping ratio of the frame. The modal damping ratio for the frame with nonlinear joint behavior and MVI effects is almost 36 times the damping ratio corresponding to the elastic joint approximation and 3.2 times the damping ratio of a nonlinear frame in which MVI is ignored. This comparison shows that elastic joint response approximation has a considerable deviation in the obtained results. Although nonlinear modeling of the connections reduces the error with respect to the elastic approach, there is still a considerable difference between dynamic characteristics of a frame with and without MVI. From this table, there is a greater difference between the two periods of vibration for the undamped system and it seems that the rotational dampers can cover the approximation about ignoring MVI.

The ratio of the connection shear demand to its capacity (DCR) is an important factor that affects dynamic response of the frame. A parametric study is carried out and the obtained results (shown in Fig. 7) are discussed in order to study the sensitivity of the frame response to its connections shear DCR.

To achieve this, the frame connections are designed with different over-strength ratios (DCRs) and an Eigensolution is made for each design. Then, the frame damper damping coefficient is altered and the frame is reanalyzed. Each design is analyzed for four damper damping coefficients ranging from 2.5×10^5 kN m s/rad. to 1.0×10^6 kN m s/rad. The obtained results are presented in Fig. 7.

From these figures, the modal damping ratio increases as the connection shear DCR increases as a general rule. The reason for this may be larger DCR tends to yield the connection earlier and as a result show a nonlinear response. Also, as the connection shear DCR increases, the rotational dampers show a more contribution on the frame response, especially when the damping coefficient is greater.

For greater rotational damper damping coefficients, the modal damping ratio increases more rapidly with an increase of the connection shear DCR. However, the period of vibration is less affected by DCR for larger damping coefficients (Fig. 7.b). It can be concluded that as the modal damping ratio increases by rotational viscous dampers, the vibration period of the frame becomes less sensitive to the connection shear DCR and MVI effects.

12. Three-storey frame response

The dynamic response of the three story frame is examined for two loading conditions (defined in table) and three types of connection model. The modal damping ratios and the period of vibrations for the three story model are shown in Table 5. As seen in this table, when the MVI is considered the vibration frequency of the system increases considerably.

From this table, the beams are loaded on three points ($F1 = F2 = 400$ kN), a sudden increase in the vibration period is achieved when the connection type alters from Type II to III. However, a relatively small difference is obtained between two connection models when the beams are loaded only at the mid span ($F1 = 0$). This shows that as the shear demand of the connection increases, the frame becomes more sensitive to MVI. To examine the impact of connection shear DCR on dynamic response of the three-storey frame, a parametric study is conducted and the obtained results are presented in Fig. 8.

The plots, shown in Fig. 8 have a similar pattern to the single-storey frame analyses results. The vibration period and modal

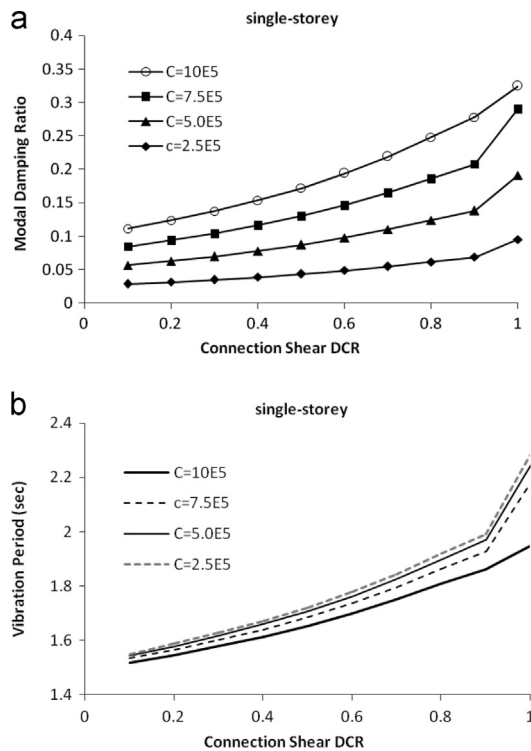


Fig. 7. a. Modal damping ratio and b. Vibration period, Fig. 7. Single-storey frame dynamic characteristics versus connection shear DCR.

Table 5

The vibration period of the 1st mode for the 3-storey frame, $c=3000(\text{kN m s/rad})$, $S_R=55 \text{ kN}$, $W_y=984 \text{ kN}$.

Connection type	F1=0.0, F2=400 kN		F1=F2=400 kN	
	ξ	T (s)	ξ	T (s)
I	0.012	3.131	0.012	3.131
II	0.129	5.058	0.125	5.042
III	0.138	5.214	0.227	6.581

damping ratio grow as the connection shear DCR and the rotational damper damping coefficient increase. As the connection shear DCR increases, the dynamic response of the system changes with a higher rate. This pattern is similar to the observed results for the single-storey frame. As the damper damping coefficient increases, the effects of connection shear DCR on the modal damping ratio becomes more visible. However, as the damper damping coefficient increases the vibration period of the frame becomes less sensitive to the connection shear DCR. These findings imply that a more reliable semi-rigid system can be achieved by using rotational viscous dampers with proper damping coefficients.

13. Time-history response analysis

Dynamic characteristic of semi-rigid frames, studied in the above sections, can mostly impact the response of these frames under dynamic loading of the seismic action. Accordingly, the seismic response of the single-storey frame is examined in this section considering MVI effects and damping. Three earthquake actions, Tabas (1978), Kobe (1995) and Duzce (1999) are used for this purpose. The records are scaled to 0.35 g. The frame is analyzed with and without MVI effects, first. Then the effect of damping in controlling the frame response is studied.

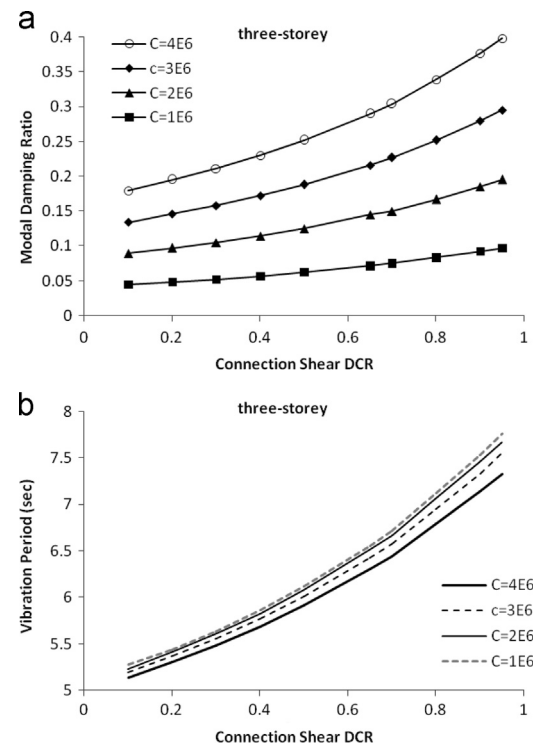


Fig. 8. a. Modal damping ratio, b. Vibration period, Fig. 8. three-storey frame dynamic characteristics versus connection shear DCR.

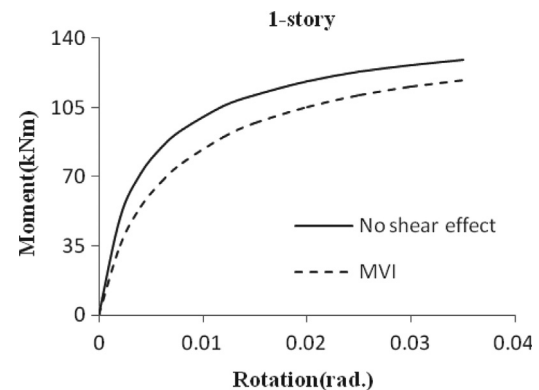


Fig. 9. Moment–rotation relations for the single-storey frame connection.

14. Analysis method

The initial gravitational load, applied as the first load case, has a larger contribution on the connection shear force comparing to the one induced by the horizontal seismic excitation. Hence, when the frame is only subjected to the horizontal component of the ground motion, the connections shear and R_{stiff} parameter can be assumed unchanged and MVI effects can be ignored during the earthquake action. This enables using a single $M-\theta$ curve during the seismic loading which is obtained immediately after applying the gravity loads by using the previously discussed FORTRAN program. Fig. 9 shows the $M-\theta$ relations implemented for the semi-rigid connections.

Since a single moment-rotation curve (independent of shear effects) is used in the analysis, the time-history analysis of the frame can be performed using any advanced FE program. Herein, ANSYS 5.4 multi-purpose FE program is used for time-history analysis [21]. The numerical modeling of the frame (including the number of nodes and elements) is the same as the modeling discussed previously. Elastic beam formulation (BEAM 3 from

ANSYS element library) is used to model the beams and columns. Nonlinear rotational response of the connections is modeled using COMBIN39 element. This joint element can capture the moment-rotation curve using up to 10 points. The rotational viscous dampers are modeled using COMBIN14 element. This joint element includes a spring with constant rotational stiffness in parallel with a damper. Since the rotational response of the connections are modeled using COMBIN39, the elastic spring coefficient of COMBIN14 is taken as zero and a constant damping coefficient is assigned for it. Lumped masses of the frames are modeled using single-node MASS21 element. Only translational values are assigned for the nodal masses. The Large Displacement capability of the software is activated to consider the $P-\Delta$ effects induced by the gravity load.

15. Frame responses

In order to better understand the importance of the MVI effects on the seismic performance of the semi-rigid frames, the exemplary single-story frame is first analyzed excluding the rotational dampers. The story deflection histories of the frame subjected to the three earthquakes are compared in Fig. 10 for two connection types. As seen in this figure, MVI has a bold role on the frame response and the lateral deflection response of the frame is underestimated when MVI is ignored.

The differences are 59%, 18% and 10% for the Tabas, Kobe and Duzce records, respectively.

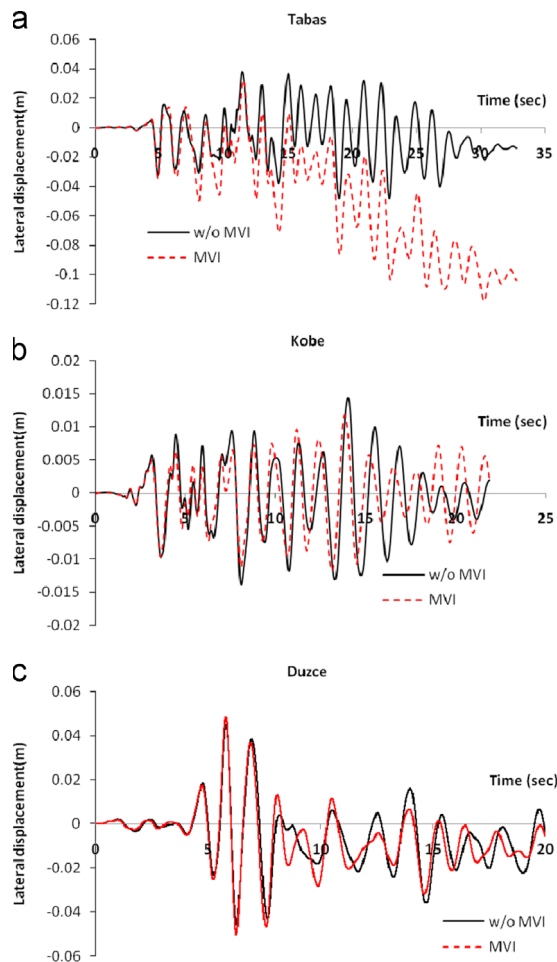


Fig. 10. Storey displacement history for the single-storey undamped frame.

The relatively less difference between two connection models under the Duzce and Kobe earthquakes with respect to the Tabas earthquake may be due to the difference in the frequency content of each earthquake record. Unlike the results obtained for the Tabas earthquake, the rotational demand of the frame connections (story displacement to height ratio) under the Kobe action is less than 0.005 rad. for the two connection models, which is a relatively elastic response for the system. This shows the considerable contribution of MVI on the frame response, even within the elastic rotational demand.

The analyses results for the frame show that when the MVI is considered, the frame sustains more lateral drift. For the case of TABAS ground motion, the single-story frame becomes almost unstable and sustains a considerable residual drift. However, ignoring MVI effects it can not be captured.

Fig. 11 shows the the left side connection shear history under the Tabas earthquake. As seen in this figure, connection shear changes in maximum by 14% during the earthquake with a negligible impact on R_{stiff} . This supports the initial assumption of constant R_{stiff} during the seismic loading.

Although there are numbers of factors affecting the frame response, the reduction of the energy dissipation capacity for the connection (the area under the moment-rotation curve) due to presence of shear force may be the main reason for the magnified frame response. Implementation of rotational viscous dampers may be an alternative to compensate this reduction. Fig. 12 compares the response of two models with and without dampers and shows the contribution of the rotational viscous dampers on reducing the frame response.

According to the plots in Fig. 12, implementation of the rotational viscous dampers can enhance the frame performance, considerably. The reduction in frame response is 48%, 33% and 9.4% for the Tabas, Kobe and Duzce earthquakes, respectively. This rate of the response reduction conforms to the increase in the lateral displacement of the frame for ignoring the MVI effects (Fig. 9). As a result, it can be concluded that rotational viscous dampers can compensate the reducing effects of MVI for the single story frame.

16. Conclusion

Nonlinear vibration response of the damped semi-rigid frames under pre-loading conditions is examined in this paper taking into account the moment-shear interaction effects. A solution algorithm is introduced for the vibration analysis of the pre-loaded semi-rigid frames considering the MVI effects of the joints. This algorithm is programmed in FORTRAN for practical frame analysis. The beams and the columns are modeled based on the traditional Euler-Bernoulli beam theory and the joints are considered as zero-length discrete elements with nonlinear moment-rotation relation. $M-\theta$ relation for this joint element is coupled with its shear force according to a previously defined formula for the bolted angle connections. The analysis starts with a zero shear force and the

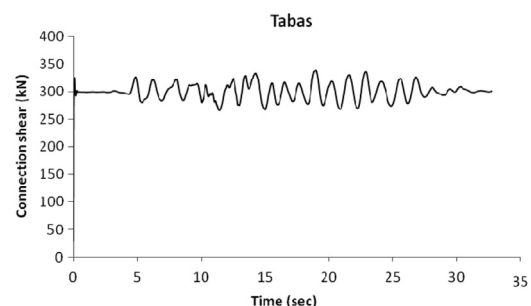


Fig. 11. Single-storey connection shear history for the Tabas earthquake.

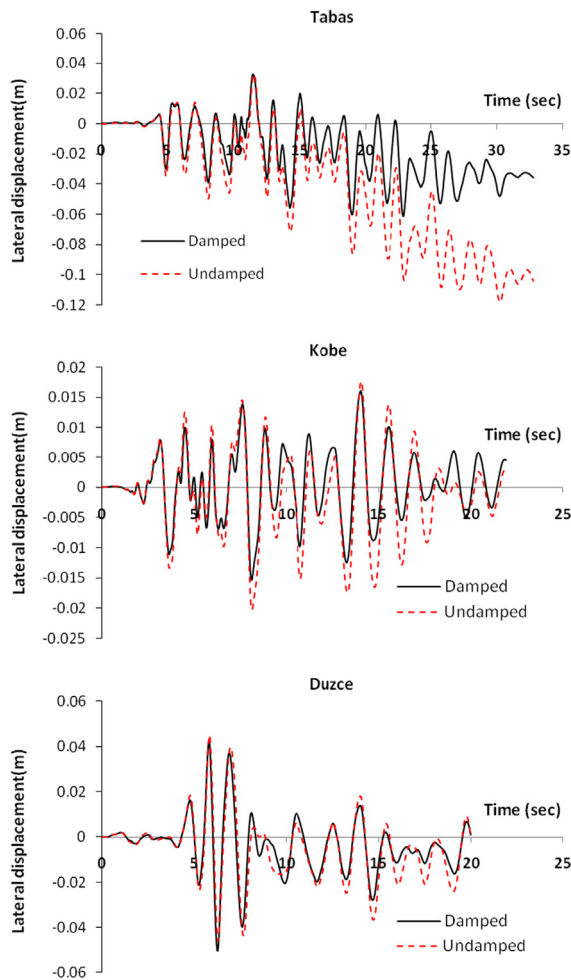


Fig. 12. Storey displacement history for the damped and undamped frame with MVI.

joint $M-\theta$ response updates iteratively until achieving a predefined accuracy. To verify the accuracy of the method, simple analytical examples are solved and the obtained results are compared with those of the FE program concluding a precise estimation and validity of the procedure. This algorithm is used to assess the vibration response of two sample frames. The main outcomes of this study are:

- The joint's shear force can affect the dynamic characteristics of the system, considerably.
- Considering MVI in modeling of the semi-rigid frames is essential when an accurate vibration analysis of these frames is aimed.
- As the connection shear DCR increases the connections become more nonlinear and MVI become more dominant.
- As the dampers damping coefficient increases the modal damping ratio of the frames become less sensitive to the MVI effects.

- Ignoring the MVI effects in the seismic analysis of the semi-rigid frames can yield considerably unsafe results.
- A more reliable semi-rigid system can be achieved by using rotational viscous dampers with proper damping coefficients.
- Implementation of rotational viscous dampers can cover the inherent error of ignoring MVI and can enhance the seismic response of the frame.

The results presented in this study are limited to the bolted angle connections. More complementary studies are needed for other types of connections such as web-angle connections or composite connections.

References

- [1] Kawashima S, Fujimoto T. Vibration analysis of frames with semi-rigid connections. *Comput Struct* 1984;19(1–2):85–92.
- [2] Chan SL. Vibration and modal analysis of steel frames with semi-rigid connections. *Eng Struct* 1994;16(1):25–31.
- [3] Chui PPT, Chan SL. Transient Response of moment-resistant steel frames with flexible and hysteretic joints. *J Constr Steel Res* 1996;39(3):221–43.
- [4] Chui PPT, Chan SL. Vibration and deflection characteristics of semi-rigid jointed frames. *Eng Struct* 1997;19(12):1001–10.
- [5] Lui EM, Lopes A. Dynamic analysis and response of semirigid flames. *Eng Struct* 1997;19(8):644–54.
- [6] Xu YL, Zhang WS. Modal analysis and seismic response of steel frames with connection dampers. *Eng Struct* 2001;23:385–96.
- [7] Sekulovic M, Salatic R, Nefovska M. Dynamic analysis of steel frames with flexible connections. *Comput Struct* 2002;80:935–55.
- [8] Sophianopoulos DS. The effect of joint flexibility on the free elastic vibration characteristics of steel plane frames. *J Const Steel Res* 2003;59:995–1008.
- [9] Pirmoz A. Evaluation of nonlinear behavior of bolted connections under dynamic loads [MS thesis]. Tehran, Iran: K. N. Toosi University; [in Farsi].2006.
- [10] A. Pirmoz Upgrading of bolted top-seat angle connections performance adding double web angles. In: *Proceedings of the 1st International Conference on Seismic Retrofitting*, Tabriz-Iran; 20–22 October: 2008. p.1–8.
- [11] da Silva JGS, de Lima LRO, da S. Vellasco PCG, de Andrade SAL, de Castro RA. Nonlinear dynamic analysis of steel portal frames with semi-rigid connections. *Eng Struct* 2008;30:2566–79.
- [12] Galvao AS, Silva ARD, Silveira RAM, Goncalves PB. Nonlinear dynamic behavior and instability of slender frames with semi-rigid connections. *Int J Mech Sci* 2010;52:1547–62.
- [13] Kishi N, Chen WF. Moment-rotation relations of semirigid connections with angles. *J Struct Eng, ASCE* 1990;116:1813–34.
- [14] Nader MN, Astanteh A. Dynamic behavior of flexible, semirigid and rigid steel frames. *J Const Steel Res* 1991;18:179–92.
- [15] Danesh F, Pirmoz A, Saedi Daryan A. Effect of shear force on the initial stiffness of top and seat angle connections with double web angles. *J Const Steel Res* 2007;63:1208–18.
- [16] Pirmoz A, Saedi Daryan A, Mazaheri A, Darbandi HE. Behavior of bolted angle connections subjected to combined shear force and moment. *J Const Steel Res* 2008;64:436–46.
- [17] Pirmoz A, Seyed Khoei A, Mohammadrezapour E, Daryan A Saedi. Moment-rotation behavior of bolted top-seat angle connections. *J Const Steel Res* 2009;65:973–84.
- [18] Pirmoz A, Danesh F, Farajkhah V. The effect of axial beam force on moment-rotation curve of top and seat angles connections. *Struct Des Spec Tall Build* 2011;20(7):767–83.
- [19] Pirmoz A. Performance of bolted angle connections in Progressive collapse of steel frames. *Struct Des Tall Spec Build* 2011;20(3):349–70.
- [20] Humar JL. Dynamics of structures 2nd ed., A. A. Balkema publishers, 2002.
- [21] ANSYS User's manual. Version 5.4.